Benchmarking Computational Methods for Cancer Cell Lineage Tracing

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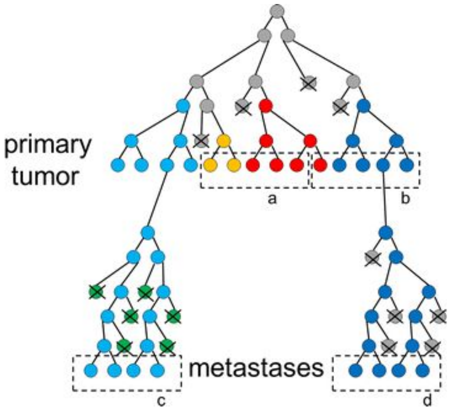
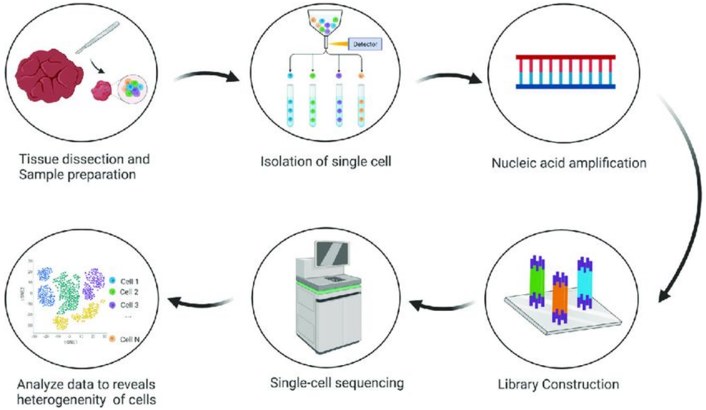
Why Study Tumor Lineages? (Biological Motivation)

- ➔ Cancer is an evolutionary process; tumors arise from a single cell and **diversify through mutations**
- ➔ Understanding **clonal evolution** helps identify treatment-resistant subpopulations and predict relapse
- ➔ Single-cell sequencing enables direct observation of heterogeneity, but **inferring evolutionary relationships** is still challenging
- ➔ Accurate lineage reconstruction reveals how driver mutations (e.g., *TP53*, *KRAS*, *PIK3CA*) shape tumor progression





Single-cell sequencing provides the raw data (mutations + expression) used to infer tumor lineage trees

Single-cell sequencing allows us to observe heterogeneity directly …
but accurate lineage reconstruction remains difficult due to dropout, noise, mutational sparsity, and copy-number alterations

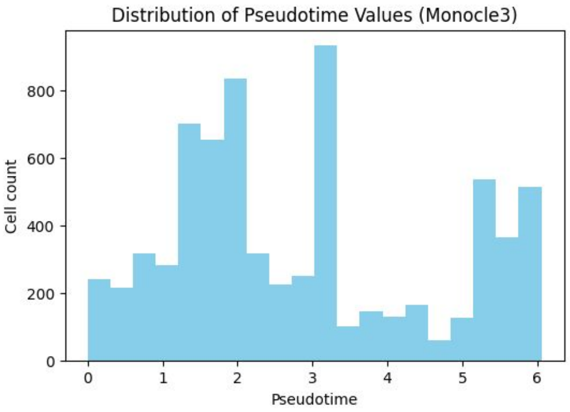
This motivates our benchmarking project:
Which computational methods best reconstruct tumor evolution from single-cell data?

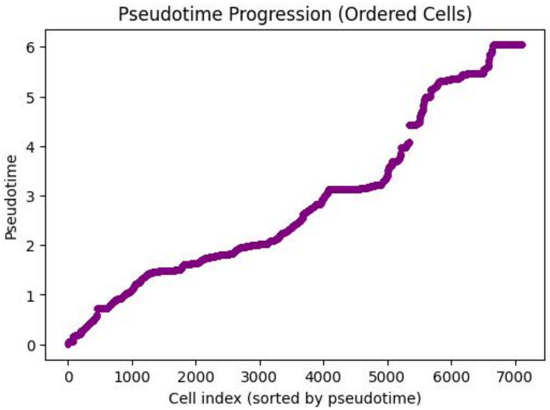
Methods Overview

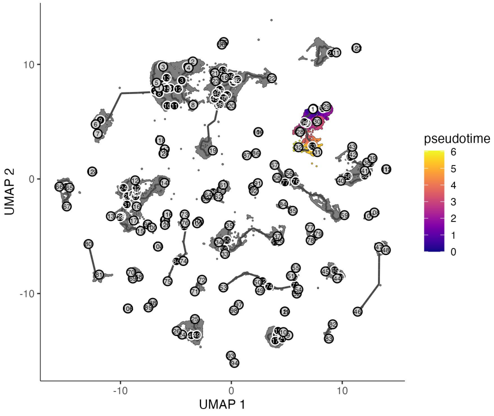
1. **Real data via Monocle3 (exploratory only)**
   - Used Monocle3 on a real breast-cancer scRNA-seq dataset to learn pseudotime and branching structure from expression, not from mutations
   - Monocle3 is a separate “biological motivation / context” result, not part of the synthetic benchmark
2. **Generate synthetic lineage data with ground truth**
   - Built three synthetic trees (balanced, mixed, comb) and corresponding cell × mutation matrices
   - For this presentation we focus on the balanced dataset as the clean benchmark
3. **Run mutation-based lineage methods on the synthetic data**
   - Evaluated on the balanced synthetic matrix: **SCITE, SiFit, PhISCS-BnB, Neighbor-Joining**
   - Additional methods were attempted, but did not produce stable or interpretable trees
4. **Comparison to ground truth**
   - For each method, we plotted its inferred tree and compared it visually to the balanced truth tree (depth, symmetry, major clades, over-/under-splitting)
   - Quantitative metrics (RF distance, ARI, edge precision/recall) are planned in next steps

Monocle3 - Real Data Results

Monocle3 reveals pseudotime structure and branching trajectories in real breast cancer data, suggesting evolving subpopulations

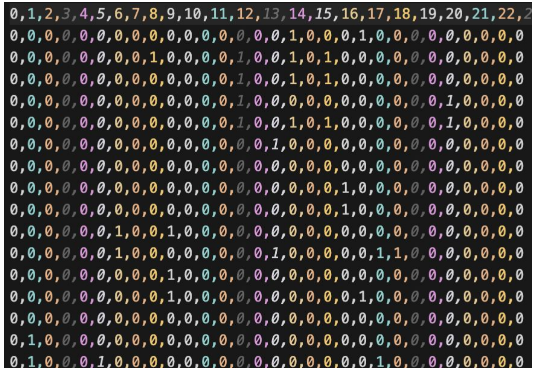






Monocle3 Findings:	**Why this matters:**
➔ Reveals clear branching structure → multiple cell subpopulations ➔ Shows a continuous pseudotime progression → potential tumor evolution path	➔ Provides biological context before running lineage algorithms ➔ Helps validate whether inferred mutation trees match real functional structure

Synthetic Dataset Generation



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Goal: Create controlled, ground-truth lineage datasets to benchmark lineage tracing algorithms

Process:

1. Build lineage tree - Generate a random tree structure, Three shapes: balanced (every split symmetrical), comb (uneven splitting), mixed

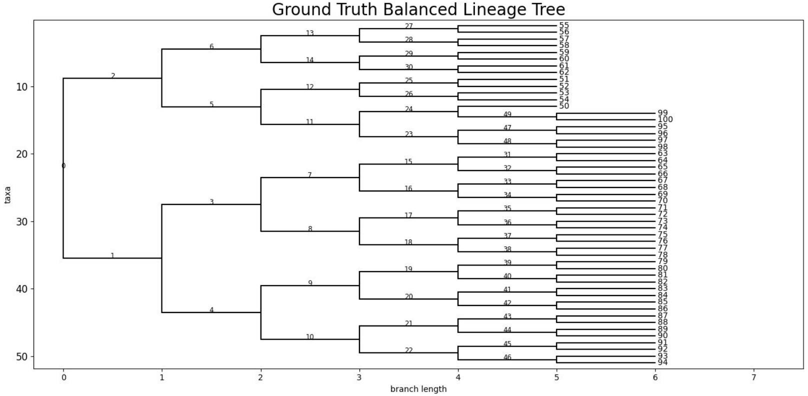
2. Generate cell x mutation matrix - Traverse each cell's path from root to leaf, collect all inherited mutations. Rows = cells, columns = mutations. 1 if mutation is present, 0 if not.

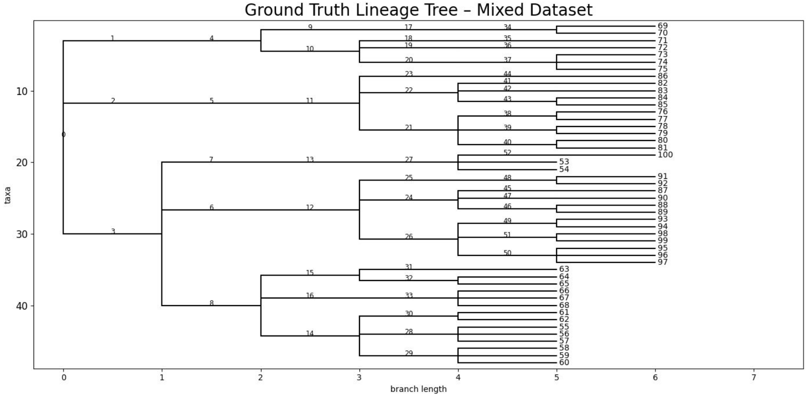
3. Group leaves into clones based on shared mutations

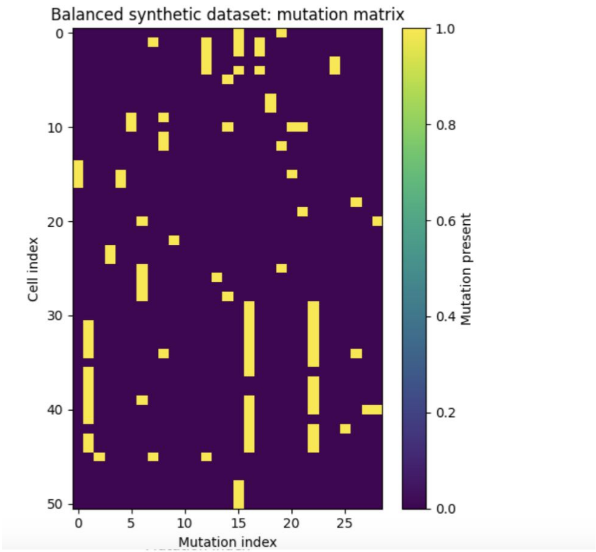
4. Apply Noise models to simulate real sequencing imperfections

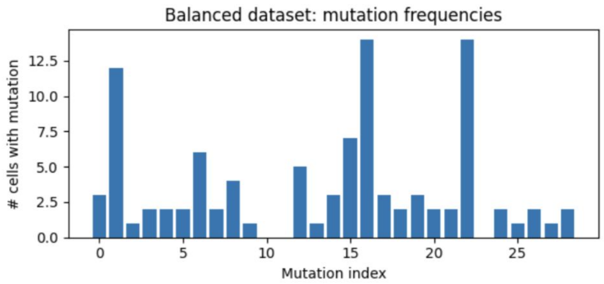
Ground-Truth Datasets for Benchmarking

Synthetic data gives us a ground truth baseline









Neighbor-Joining

Goal: Reconstruct a lineage tree using a simple, fast distance-based approach.

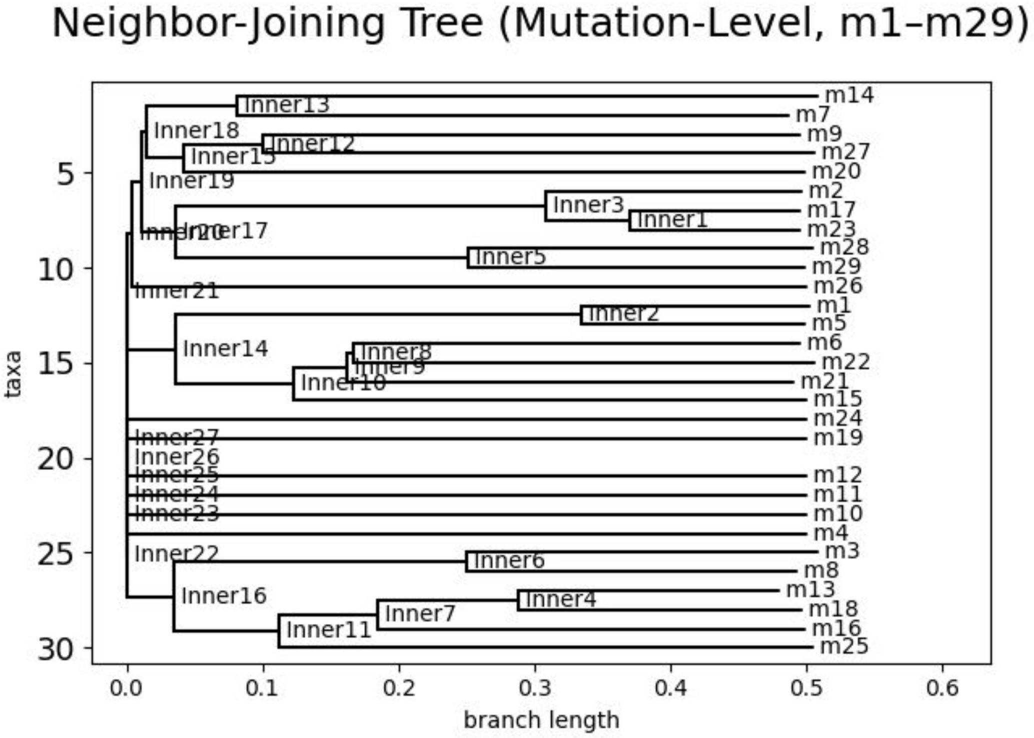
Inputs:

- Binary cell × mutation matrix
- Pairwise distance matrix derived from mutation profiles
    - (e.g., Jaccard distance on mutation presence/absence)

Neighbor-Joining (NJ) is a fast, distance-based phylogenetic algorithm that constructs a tree by iteratively joining the closest pairs of nodes based on mutation-derived distances. It makes no explicit evolutionary or noise assumptions and relies solely on similarity between samples. Because of its simplicity and speed, NJ is used as a **baseline** to contextualize more sophisticated tumor-aware lineage methods. The output is a single distance-based lineage tree in Newick format.

Neighbor-Joining Results

NJ serves as a fast, intuitive mutation-level reference tree that anchors our comparisons to more complex lineage-inference tools



Dataset: balanced synthetic dataset (51 cells, 29 mutations)

Method:

- Input: binary mutation matrix
- Distance metric: Jaccard distance between mutations
- Output: mutation-level tree (labels: m1–m29)

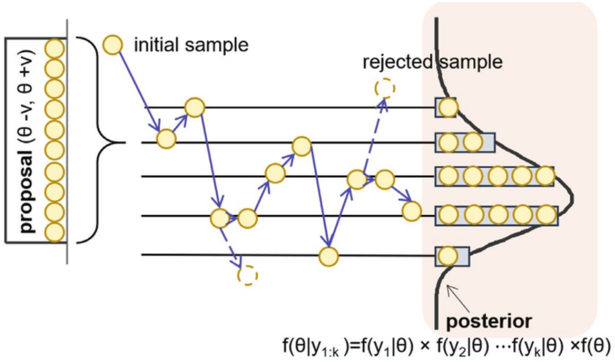
Runtime: < 1 seconds

SCITE

Goal: Reconstruct cell lineage trees from noisy single-cell mutation data.

Inputs:

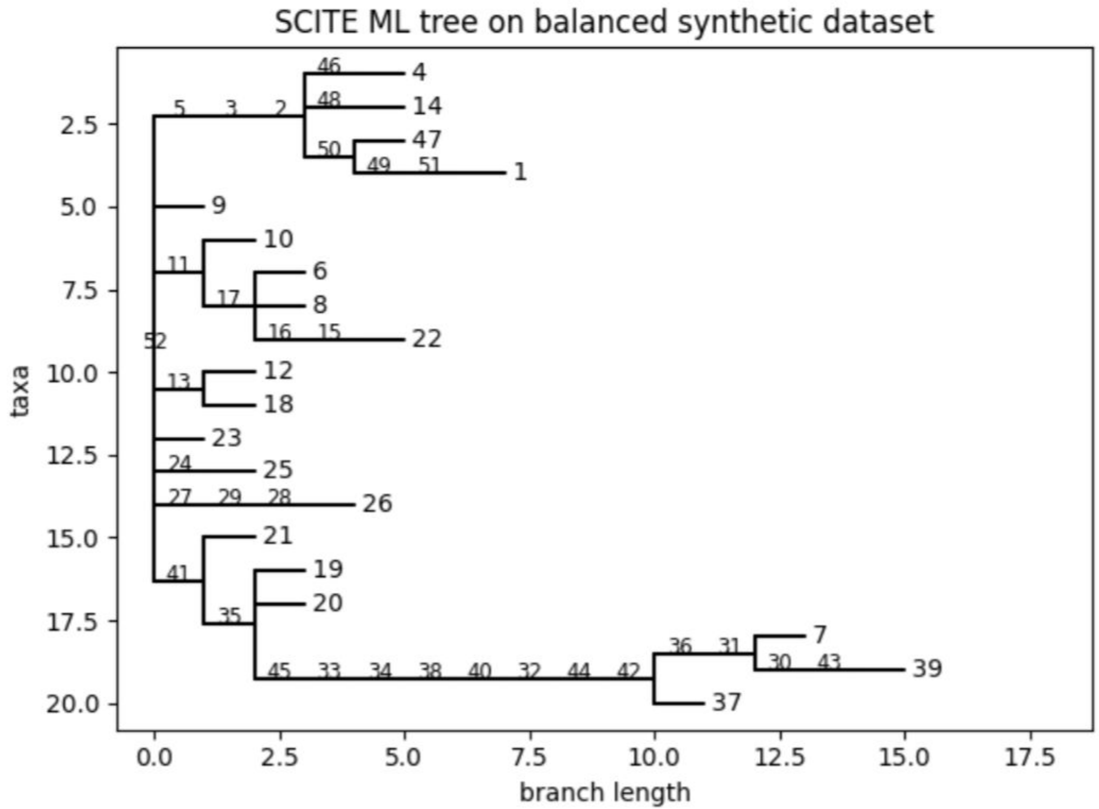
- Binary cell × mutation matrix
- Estimated sequencing error rates:
    - α = false positive rate
    - β = allelic dropout / false negative rate



SCITE formulates cell lineage reconstruction as a **probabilistic inference problem** over mutation events and tree structures. Given a binary cell-by-mutation matrix, it uses **maximum likelihood estimation with MCMC** sampling to infer mutation order and lineage topology while explicitly modeling **single-cell sequencing** errors such as false positives and allelic dropout. The method outputs **maximum-likelihood lineage trees** in Newick format along with **posterior samples** over mutation histories, enabling uncertainty assessment, performance benchmarking, and downstream comparison to ground-truth lineages.

SCITE Results

SCITE recovers major branches but may miss fine structure



Dataset: balanced synthetic dataset (51 cells, 29 mutations)

SCITE settings:

- False negative rate β = 0.10
- False positive rate α = 0.05
- Maximum likelihood (ML) tree from SCITE

Runtime: ~193 seconds on this dataset

Interpretation:

- SCITE recovers a branching clonal structure consistent with the “balanced” design of the synthetic tree
- This gives us a baseline lineage for comparison against PhyloWGS and other methods

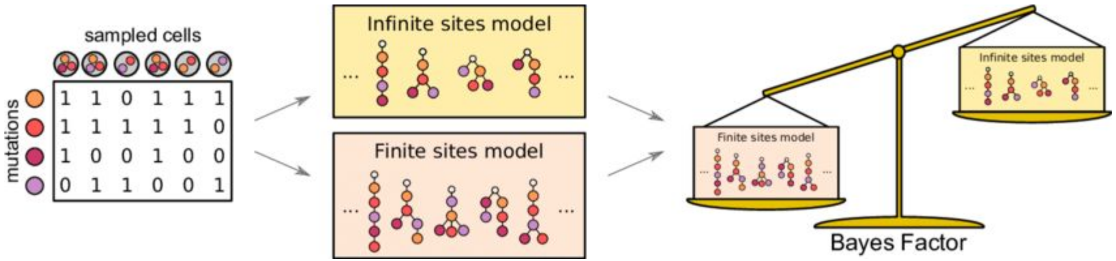
Phylogeny Inferred Using SiFit (Finite-Sites Model)

Goal: Reconstruct tumor lineage trees from noisy single-cell mutation data under a biologically realistic mutation model.

SiFit reconstructs tumor lineages using a **maximum-likelihood framework** under a finite-sites mutation model, allowing mutations to recur or be lost while explicitly modeling sequencing errors. By relaxing the infinite-sites assumption made by tools like SCITE, SiFit captures more biologically realistic mutation processes, often resulting in **deeper and more complex lineage trees**.

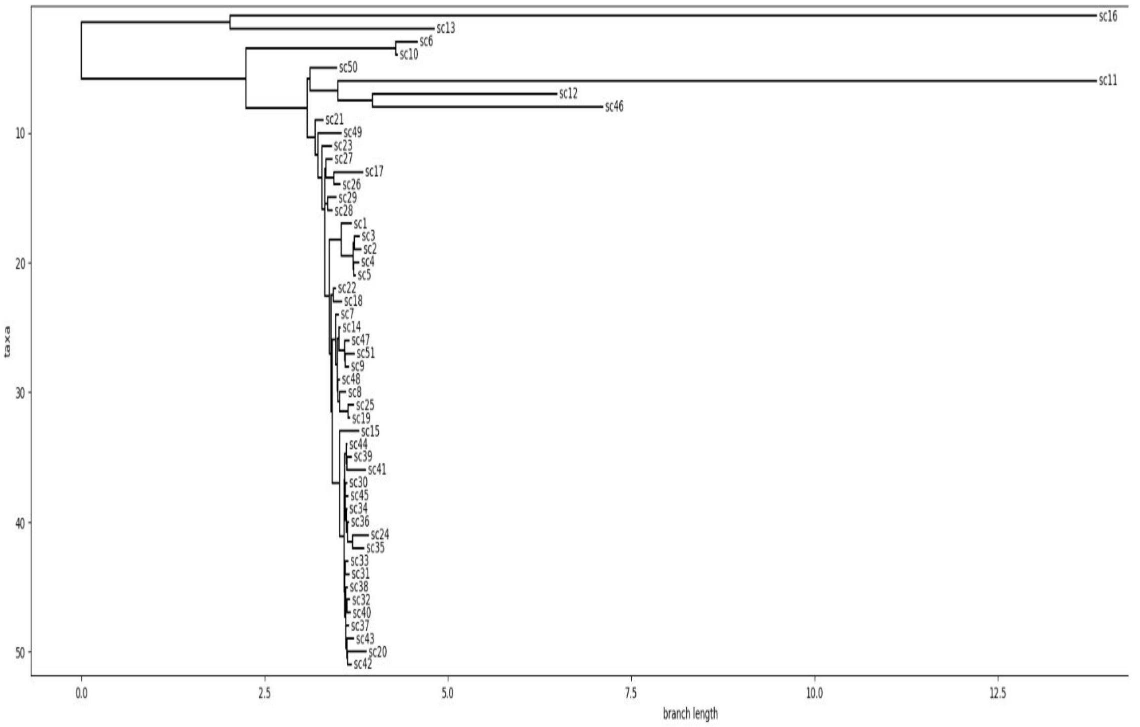
Inputs:

- Binary cell × mutation matrix
- Estimated sequencing error parameters
    - False positive rate
    - False negative (dropout) rate



SiFit Results

SiFit offers a finite-sites perspective on the same dataset, revealing mutational patterns that infinite-sites tools may miss



Dataset: balanced synthetic dataset (51 cells, 29 mutations)
SiFit settings:

- False negative rate (FN) = 0.10
- False positive rate (FP) = 0.01
- Finite-sites evolutionary model
- 10,000 iterations (maximum likelihood search)
- Binary mode (-df 0, required for SC matrices)

Output: Maximum likelihood mutation tree
Runtime: ~7 seconds
Interpretation:

- SiFit reveals deeper, more uneven branching consistent with real tumor subclonal diversity rather than the clean symmetry seen in synthetic data
- Long isolated branches highlight heterogeneous mutation burdens, suggesting biologically distinct subpopulations rather than uniform clonal expansion.

PhISCS-BnB

Goal: Infer a clonal lineage tree that satisfies perfect-phylogeny constraints from single-cell mutation data.

Inputs: Binary cell × mutation matrix (cells × mutations)

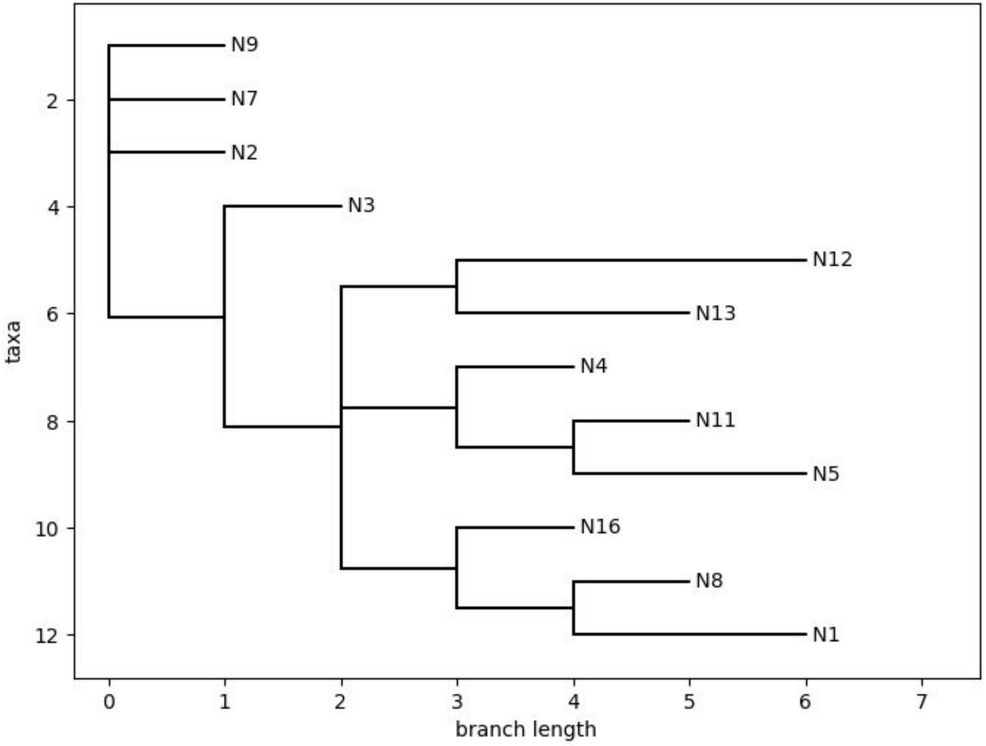
PhISCS-BnB reconstructs tumor phylogenies by formulating lineage inference as an **integer linear programming (ILP)** problem solved via **branch-and-bound optimization**. It enforces a **perfect-phylogeny model**, where each mutation arises once and is never lost, and finds an optimal solution under these constraints rather than relying on probabilistic sampling. The method outputs a **clonal lineage tree in Newick format**, optionally with mutations annotated on branches, but can be sensitive to noise and sparse mutation signals due to its strict phylogenetic assumptions.

Algorithm 1 PhISCS-BnB

Input: $I \in \{0,\ 1\}^{n\times m}$: Original input to the algorithm
Output: $Y \in \{0,\ 1\}^{n\times m}$ such that
$Y = \text{argmin}_{PP_{(y)}} F_{0\to1}(I, Y)$.

$\text{Best}_{\text{Node}} \leftarrow$ A simple PP solution (see Section 3.3)
$Q \leftarrow$ An empty priority queue
Push I in Q
while Q is not empty **do**
  $C \leftarrow$ pop next node from Q with the lowest priority score
  $\text{New}_{\text{node}_1} \leftarrow C$
  for $r \in R_{0,1}(p, q)$ **do** ▷ See Lemma 3.1
    $\text{New}_{\text{node}_1}(r, p) \leftarrow 1$ ▷ $F_{0\to1}(I, \text{New}_{\text{nod}}e_1)$ is also updated here
  $\text{New}_{\text{node}_2} \leftarrow C$
  for $r \in R_{1,0}(p, q)$ **do**
    $\text{New}_{\text{node}_2}(r, q) \leftarrow 1$
  for $i \in \{1, 2\}$ **do**
    if $\text{New}_{\text{node}_i}$ is a PP **then** ▷ i.e. $\text{New}_{\text{node}_i}$ is a leaf
      if $F_{0\to1}(I, \text{New}_{\text{node}}i) < F_{0\to1}(I, \text{Best}_{\text{Node}})$ **then**
        $\text{Best}_{\text{Node} \leftarrow \text{New}_{\text{node}_i}}$
    else ▷ Use any bounding algorithm proposed in Section 3.2
      $\text{lb} \leftarrow$ A lower bound for the number of flips to get to a PP matrix from $\text{New}_{\text{node}}i$
      if $\text{lb} + F_{0\to1}(I, \text{New}_{\text{node}}i) < F_{0\to1}(I, \text{Best}_{\text{Node}})$ **then**
        Push $\text{New}_{\text{node}}i$ in Q with priority score set to $\text{lb} + F_{0\to1}(I, \text{New}_{\text{node}}i)$
Return $\text{Best}_{\text{Node}}$

PhISCS-BnB Results



Dataset: balanced synthetic dataset (51 cells, 29 mutations)
SCITE settings:

- Branching factor b = 2
- Draw tree t = 1

Runtime: ~53 seconds on this dataset
Interpretation:

- Unlike SCITE, requires every mutation occurs once and does not revert - causes shallow tree, limited branching
- Over-simplify lineage structure when the dataset contains noise

Comparison of Mutation-Based Lineage Inference Methods

They all start from the same single-cell mutation matrix, but they make *very different assumptions*, which is why their trees diverge

Feature	SCITE	SiFit	PhISCS-BnB	Neighbor-Joining
Primary data	Single-cell mutation matrix	Single-cell mutation matrix	Single-cell mutation matrix	Mutation-derived distances
Evolutionary model	Infinite sites	Finite sites (allows repeated mutations)	Perfect phylogeny (ILP)	None (distance-based)
Handles dropout/noise	Yes (α, β model)	Yes (explicit error model)	Limited (constraint-based)	No explicit noise modeling
Output structure	Single ML cell lineage tree	Single full mutation tree	Clonal tree (mutations on edges)	Tree based on similarity
Computational cost	Moderate (MCMC)	Fast	Moderate	Very fast
Strength	Robust to scRNA-seq noise	Biologically realistic mutation model	Guarantees phylogenetic consistency	Simple, assumption-free baseline
Limitation	Infinite-sites assumption	Can produce deep/complex trees	Sensitive to noise/constraints	Not tumor-aware
Observed in our project	Recovers some balanced structure but collapses branches under conflict	Produces rich structure on real tumor data; captures heterogeneity	Over-simplifies synthetic balanced tree into few large clones	Mismatches ground truth; non-biological bushy shape

Methods Not Aligned to Our Synthetic Setup

PhyloWGS:

- Designed for bulk sequencing VAFs
- Our binary single-cell genotypes violate its assumptions → flat posterior, no consensus tree

LinTIMaT:

- Requires mutation + expression matrices
- Mutation-only data breaks its model → cannot infer trajectories

OncoNEM:

- Assumes infinite-sites and low noise
- Our noisy mutation model makes it return oversimplified trees

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Exception: 85% of trees are multiprimary (1700 of 2000),
 so not enough to report good posterior.
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Next Steps

- Apply SCITE, SiFit, PhISCS-BnB, and Neighbor-Joining to a **real single-cell tumor mutation dataset**, verify conclusions from synthetic datasets

- Compare real-data lineage structures to synthetic benchmark behavior, create **numeric ranking method** to compare trees

- **Aggregate results** and draw biological interpretation running all four methods in a single real dataset

- Summarize tradeoffs to **provide guidance on method selection** for different data types

Citations

- Jahn et al., SCITE: Inferring tumor evolution from single-cell sequencing, Bioinformatics (2016)
- Morrissy et al., Single-cell characterization of medulloblastoma reveals therapy-induced stemness, Nature (2024)
- Satas & Raphaël, Accurate, fast, and memory-efficient inference of single-cell phylogenies, Genome Research (2020)
- Ogawa et al., Computer-assisted DNA phylogeny reconstruction based on restriction enzyme cleavage patterns, Gene (1987)